THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The principle of conservation of energy is an example of: 1

(a) a broad idea that helps make sense of many situations
(b) a unit of energy
(c) a local rule of a laboratory
(d) a measuring instrument

2. If one breath is about 0.5 litre and there are about 20 000 breaths a day, the air breathed in a day is about: 1

(a) 10 000 litres **(b)** 1 lakh litres **(c)** 100 litres **(d)** 1000 litres

3. 2 g of a solute is dissolved in water to make 50 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 2 % **(b)** 3.85 % **(c)** 8 % **(d)** 4 %

4. In a solution of sugar in water, the solvent is: 1

(a) sugar **(b)** water **(c)** both **(d)** neither

5. Carbon monoxide has C : O = 3 : 4. The mass of oxygen that combines with 9 g carbon is: 1

(a) 9 g **(b)** 6 g **(c)** 16 g **(d)** 12 g

6. In a nitrogen molecule the number of shared electron pairs is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

7. The isotope used in the treatment of goitre is: 1

(a) uranium-235 **(b)** cobalt-60 **(c)** carbon-14 **(d)** iodine-131

8. A cyclist covers 1500 m in 100 s along a straight road without turning back. The average speed is: 1

(a) 15 m s^{-1} **(b)** 0.07 m s^{-1} **(c)** 7.5 m s^{-1} **(d)** 30 m s^{-1}

9. The acceleration of a freely falling body near Earth's surface is about: 1

(a) 1.6 m s^{-2} **(b)** 9.8 m s^{-2} **(c)** 98 m s^{-2} **(d)** 0.98 m s^{-2}

10. Galileo's thought experiment on a smooth horizontal plane suggested that a body: 1

(a) stops immediately
(b) always needs a force to move
(c) keeps moving indefinitely if no friction acts
(d) moves in a circle

11. We can walk because the ground exerts a _____ on our feet in the forward direction. 1

(a) force of friction **(b)** magnetic force
(c) tension **(d)** buoyant force

12. 1 kilowatt equals: 1

(a) 10 000 W **(b)** 100 W **(c)** 1000 W **(d)** 10 W

13. In a sound wave, a region of higher density is called a: 1

(a) trough **(b)** crest of a transverse wave
(c) rarefaction **(d)** compression

14. The colourless plastids that store starch, oil or proteins are: 1

(a) mitochondria **(b)** chloroplasts **(c)** leucoplasts **(d)** chromoplasts

15. Mitochondria and plastids resemble bacteria because they have their own: 1

(a) nucleus **(b)** cell wall
(c) DNA and ribosomes **(d)** cell membrane only

16. The joint at the knee and elbow is a: 1

(a) ball and socket joint **(b)** hinge joint
(c) pivot joint **(d)** fixed joint

17. Tapeworm and planaria are: 1

(a) flatworms **(b)** molluscs **(c)** annelids **(d)** roundworms

18. Which is a sexually transmitted infection? 1

(a) cholera **(b)** typhoid **(c)** malaria **(d)** HIV/AIDS

19. The technique in which fertilisation happens in a laboratory dish is: 1

(a) layering **(b)** budding **(c)** IVF **(d)** grafting

20. Water enters the atmosphere from plants through: 1

(a) transpiration **(b)** infiltration **(c)** precipitation **(d)** condensation

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. What is a prediction in science and how is it different from a guess? 2

22. Why does the law of definite proportions apply to compounds and not to mixtures? 2

23. Define average velocity. State its SI unit. 2

24. State Newton's first law of motion. 2

25. Define osmosis. 2

26. What are meristematic tissues? State where they are found. 2

27. What is vegetative propagation? Give two examples. 2

28. Differentiate between the solar constant and insolation. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Describe how a separating funnel separates two immiscible liquids. Name the property used. 3

MASTERMIND

30. Explain with the examples of sodium, oxygen and carbon how the valency of an element is determined. 3

31. Why is climbing an inclined plank easier than lifting a load vertically? Does it save work? 3

32. A parking sensor emits ultrasound that is reflected from an obstacle 1.2 m away (v = 345 m s^{-1}). Find the time taken for the round trip. 3

33. Distinguish between voluntary and involuntary muscles with examples. 3

34. Why have arthropods been so successful? 3

35. In a closed vessel 4.0 g $CaCO_3$ and 2.92 g HCl react to give 1.76 g CO_2, 0.72 g H_2O and some $CaCl_2$. Find the mass of $CaCl_2$ formed. 3

OR

Explain the formation of sodium chloride from its atoms. 3

36. Explain how deforestation affects the water cycle. 3

OR

Suggest three actions to restore balance among the Earth's spheres. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Write the chemical formulae of calcium chloride, aluminium oxide, sodium sulfate, magnesium oxide and ammonia. 5
(ii) Name the rule used.
(iii) Name CO, CO_2, CS_2, N_2O_4, PCl_3.

OR

(i) What is a molecule? 5
(ii) Distinguish between a molecule of an element and of a compound with two examples each.
(iii) Why does helium exist as single atoms?

38. **Case study:** A weightlifter holds a barbell of total mass 30 kg (g = 9.8 m s^{-2}). 5
(i) What forces act on it?
(ii) Find the weight.
(iii) Find the force she applies.
(iv) What would be the net force?

OR

(i) Explain the system approach for two boxes of m_1 and m_2 pulled by force F. 5
(ii) Find the acceleration for m_1 = 2 kg, m_2 = 3 kg, F = 10 N.
(iii) Find the tension.

39. **Case study:** Onion root tip cells show different shapes of chromatin under the microscope. 5
(i) Why are root tips used?
(ii) What process is seen?
(iii) Name its types.
(iv) What is the use of this process in the body?

OR

(i) What is the cell cycle? 5
(ii) Compare mitosis and meiosis.
(iii) State the importance of cell division.

40. (i) Describe the human male and female reproductive systems. 5
(ii) State the functions of the testes and ovaries.

OR

Case study: Ravi is growing taller, his voice is changing and his shoulders broaden. 5
(i) Which stage is this?
(ii) What brings about these changes?
(iii) Mention two changes in girls.
(iv) Why is a balanced diet important?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND